Kernix baseline-0.1.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

05 workloads

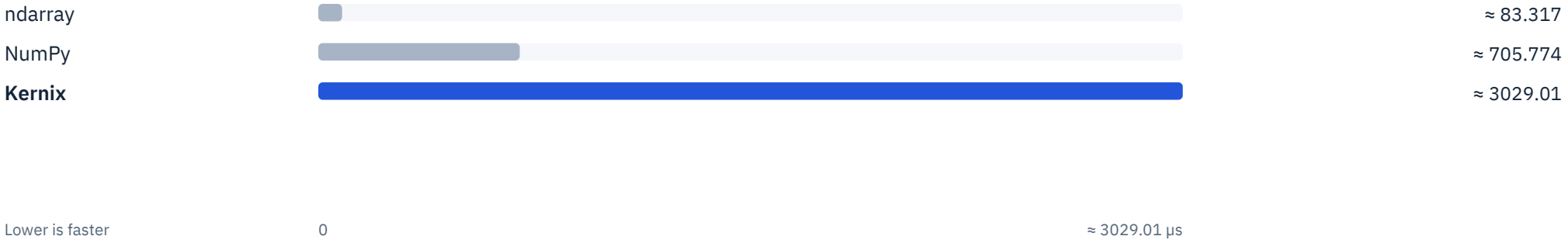
15 observations

Source precision retained

Matrix multiplication

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Historical medians of 15 batch means. Recorded p95 is the batch maximum.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix baseline-0.1.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Matrix multiplication

Batch-mean median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product

Kernix baseline-0.1.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Elementwise addition

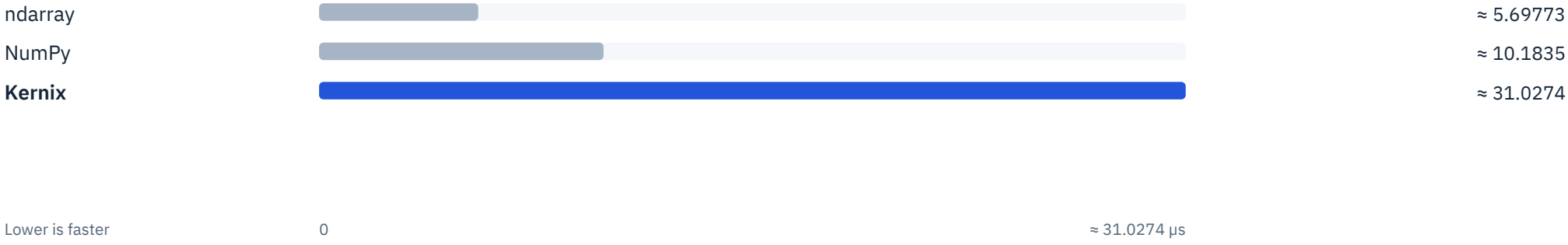
65,536 elements · f64 · 1 worker · allocated output

Batch-mean median

Sum reduction

65,536 elements · f64 · 1 worker · scalar output

Batch-mean median



Kernix baseline-0.1.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Batch-mean median

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	32012.596670934698	32.012596670934698
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1502.201405996515	1.502201405996515
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	11043.529801324503	11.043529801324503
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	3029011.111111111	3029.011111111111
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	83316.96677740864	83.31696677740864
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	705774.3333333334	705.7743333333334
Elementwise addition / Kernix 65,536 elements · f64	23401.578110383536	23.401578110383536
Elementwise addition / ndarray 65,536 elements · f64	21198.666101694915	21.198666101694915
Elementwise addition / NumPy 65,536 elements · f64	23480.653521126762	23.480653521126762
Sum reduction / Kernix 65,536 elements · f64	31027.383374689827	31.027383374689827

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sum reduction / ndarray 65,536 elements · f64	5697.730856882406	5.697730856882406
Sum reduction / NumPy 65,536 elements · f64	10183.508957654723	10.183508957654723
Row softmax / Kernix 256 × 64 · f64	397787.5396825397	397.7875396825397
Row softmax / ndarray 256 × 64 · f64	67230.14516129032	67.23014516129032
Row softmax / NumPy 256 × 64 · f64	94601.70943396227	94.60170943396227